Perception by Generation: Evaluating Emergent Robotic Understanding in Video World Models

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1 Background

1.1 Motivation

The field of embodied AI and computer vision is shifting away from task-specific specialist architectures such as (Carion et al. 2026; Lin et al. 2025; J. Wang et al. 2025). The self-supervised alternatives that replace them each trade off along one axis: masked autoencoding (Limin Wang et al. 2023; Zhang et al. 2025) preserves geometry and motion but generalises weakly to semantics, contrastive and language-aligned methods (Radford et al. 2021; Tschannen et al. 2025; Siméoni et al. 2025) give semantics but discard spatio-temporal precision, and latent prediction (Assran et al. 2025) is broadly strong across both at the cost of a complex pretraining recipe and no generative capability.

Instead, text-to-video diffusion backbones are seeing increased usage as comprehensive world models. Large-scale video generation is thought of not merely as a synthesis objective but a pre-training paradigm for visual perception. In order to produce temporally and geometrically consistent video, a model must internalise 3D structure, object permanence, and physical interaction. Recent literature demonstrates that these models inherently encode robust spatio-temporal priors, geometry, and semantics, enabling them to match or exceed dedicated networks across diverse perception tasks through representation probing (King et al. 2026) or post-training (Letian Wang et al. 2026).

However, there is a training-free alternative that leverages direct generative readout, prompting models like Veo3 directly with tasks requiring geometry and physics understanding and reading the answer off the generated frames (Wiedemer et al. 2025). This generative approach uniquely supports predictive temporal reasoning, and it remains viable for evaluating closed-source frontier models where weights-access is limited.

1.2 Research Gap

Embodied AI currently divides research across two model families: Vision-Language-Action models (VLAs) based on Vision-Language Models (VLMs), valued for the physical priors they inherit from internet-scale multimodal pretraining, and World Action Models (WAMs) based on video generation models, used as predictive simulators for policy learning, policy evaluation, and synthetic data generation. Evaluation practice has followed this division, and both of its branches (VLM embodied question answering validated against real robot execution (Xie et al. 2026), and trustworthiness properties of WAM generated rollouts (H. Li et al. 2026)) score a model through the outcome it produces. A video world model's underlying representation of the robot scene, its geometry, object state, and physical dynamics, is therefore measured only as filtered through a task-level verdict, which a model can reach correctly over a scene it has

rendered incoherently, and fail on despite rendering the scene faithfully. What the generated frames actually assert about the scene is not scored.

The representation-level alternatives inherit their own confound. A trained probe has capacity of its own, so a positive result establishes only that the information is linearly decodable from the backbone, not that the backbone uses it; probe training on in-domain robot data compounds this by allowing the probe to fit regularities the frozen features never encoded. Generative readout avoids both, but it has so far been measured almost entirely on generic web video, on physics demonstrations, geometry puzzles, and visual analogies. The properties that determine whether a world model is usable as a robot simulator remain unmeasured: the semantic and geometric layout of the workspace, which object an instruction refers to and where it affords a grasp, gripper–object contact state, and the configuration of the robot arm itself. Existing readout results, such as those in Wiedemer et al. 2025, are also largely qualitative demonstrations rather than systematic evaluations with defined scoring metrics.

This absence carries a cost, because the generated frames are themselves the consumed artifact: as synthetic training data, as predictive rollouts, and, in generative readout, as the medium in which the answer is read. If contact, occlusion, or support are systematically wrong while the frames remain visually plausible, that error propagates into whatever consumes them, and aggregate task metrics will not surface it. Direct generative readout targets this gap by querying what the model represents about the robot scene in the frames themselves.

2 Related Work

2.1 Perception from Video Generation

Video generation models contain latent visual understanding that can be surfaced with little or no supervision. Few-shot LoRA adaptation on input–output transition videos displays the ability to perform a range of perception tasks (Acuaviva et al. 2025). Frontier generators have been prompted, with no task-specific training or fine-tuning, to perform segmentation and edge detection and to exhibit early step-by-step visual reasoning across frames (Wiedemer et al. 2025). Two recent lines of work convert this latent capability into a usable interface. *GenCeption* turns a pre-trained video diffusion backbone into a single-step feed-forward perception model through multi-task post-training (Letian Wang et al. 2026). *Gen4U* approaches it from the representation side, probing the intermediate activations of Veo 3 and Wan2.2 with mutual k-nearest-neighbour alignment metrics to identify a depth and noise level from which a single forward pass yields features competitive with dedicated video encoders on semantic and geometric tasks, and finding that alignment quality scales with generative capability (King et al. 2026). Both routes require weight access, which is not available to external researchers on the frontier closed models; the prompting-only evidence that does apply to them remains largely qualitative, establishing that perception emerges without quantifying how far it extends. We measure that extent directly, treating the generator as a frozen black box and prompting as the only interface.

2.2 Video Generation Benchmarking

Evaluation of video generation has matured along a stable set of axes: visual quality, temporal coherence, text–video alignment, and physical plausibility. Established protocols such as *VBench*, *VideoPhy*, *WorldModelBench* combine automatic perceptual metrics with human or MLLM judgement of whether generated dynamics are physically admissible (Huang et al. 2023; Bansal et al. 2024; D. Li et al. 2025). These axes were designed for open-domain synthesis, where the question is whether a video looks and behaves like plausible footage, and where the video is therefore evaluated as an artifact. We treat it instead as a readout: conditioned on a real

robot observation, the content of the generated video is an assertion about that scene, and the correctness of the assertion is orthogonal to how good the video looks.

2.3 Embodied and Robotic Benchmarking

As video world models moved into robotics, these axes followed. *EWMBench*, *WorldArena*, and *RoboWM-Bench* assess scene stability, motion plausibility, action completeness, and physical executability of generated manipulation video (Yue et al. 2025; Shang et al. 2026; Jiang et al. 2026). *RoboTrustBench* pushes furthest along this line, evaluating seven video world models on expert-validated instruction–image pairs from DROID under normal, constraint-sensitive, counterfactual, and adversarial conditions, and finding that visually coherent generation coexists with failures in constraint reasoning, counterfactual grounding, and physical interaction (H. Li et al. 2026). Common to this family is the object of study: each evaluates a predicted future, asking whether the video a model generates is realistic and faithful to the instruction, rather than whether the model has correctly parsed the scene it was given. *RoboVista* asks the latter question, but of a different model class, decomposing real robot pipelines into perception, high-level decision making, motion estimation, and failure recovery and evaluating VLAs on expert-annotated visual question answering across task types; notably, benchmark accuracy there correlates with physical task success (Xie et al. 2026), suggesting that perceptual benchmarks of this kind carry downstream signal, at least for models that answer in language. We ask the generator itself to render what it perceives in the scene it was given, as dense pixel-space output scored against specialist references instead of an answer key.

3 Goal and Objectives

Goal To determine, systematically and quantitatively, what video world models understand about robot scenes by prompting them to render that understanding directly into pixels, and to characterise where and how that understanding breaks down. The intent is not to compete with task-specific perception models, which will remain more accurate on the tasks they were built for, but to measure whether the structure a robot simulator must possess is present at all, how it degrades across scene types and scenario conditions relative to specialist and language-output baselines, and how it fails.

The goal decomposes into four objectives:

O1 — Property taxonomy and readout feasibility.
Define the set of scene properties to be measured. The current shortlisted list of perception tasks would be: semantic segmentation (objects and embodiment), depth perception, robot keypoint detection, contact detection, grasp region segmentation, robot pose estimation, active object detection.

O2 — A scoring protocol for generative readout.
Construct the procedure that converts a rendered rollout into a quantitative measurement, for each of the listed perception tasks.

O3 — Quantitative evaluation across models and domains.
Apply the protocol to open-weight and frontier closed generators over real robot observations, and report the resulting scores against specialist references and ground truth, depending on the perception task.

O4 — Failure characterisation.
Produce a taxonomy of how generative readout fails, since the failure modes available to a model answering in pixels are structurally unlike those available to one answering in language, and no such taxonomy exists.

4 Research Plan

4.1 Phases

1. **Scoping and literature consolidation.**
   Conduct comprehensive literature review on the three domains that the proposal overlaps: perception from video generation, video generation benchmarking, and embodied benchmarking. Decide on a preliminary shortlist of perception tasks to measure (per above in O1), further look into the background and best practices associated with the rigorous evaluation of those tasks. Develop a working local generation environment, validated by reproducing a published qualitative readout result (for example, from Wiedemer et al. 2025) end to end, so that the pipeline is known to be good before any further detailed measurements depend on it.
2. **Data curation.**
   The evaluation is built on RoboTrustBench (H. Li et al. 2026): 1,207 expert-validated instruction–image pairs from DROID (Khazatsky et al. 2025), stratified over 10 scene types, 321 object types, and 102 task verbs. It is chosen because its item format is already the conditioning format this work requires: a single real observation frame with the robot embodiment visible, paired with a language instruction. Its scenario axis (Normal, Constraint-Sensitive, Counterfactual, Adversarial) is repurposed as a stress dimension: the original benchmark asks whether a continuation is trustworthy, we ask whether the model can report what it sees, and the counterfactual items are the natural probe for scene substitution.
   A 20-item pilot subset is drawn proportionally to the scenario distribution (Table 1), with a floor of two Adversarial items so that both subcategories are covered. Within each cell, sampling is seeded, secondarily stratified over scene and object category, and constrained to one item per DROID episode. Pilot items are flagged and retained in the main run, with a pilot/non-pilot sensitivity check reported.
3. **Task taxonomy and feasibility pilot.**
   Each of the seven shortlisted tasks is specified as a prompt template with a defined output convention and piloted on the 20-item subset. The present task set measures static scene properties; predictive readout is left to future work. The tasks are organised into three hierarchical levels of understanding (Table 2):

   1. Scene — What entities and geometries exist in the scene?
   2. Interaction — How can the robot interact with the entities?
   3. Robot State — What configuration is the embodiment itself in?

   The pilot determines the feasibility and efficacy of results extraction for each task, starting with the tasks where the anchors calibrate expectations against published results from specialist reference models. After the pilot, the task set, prompt templates, and ground-truth sources are frozen.
4. **Evaluation protocol.**
   The item set is converted into a frozen scoring pipeline, according to the choices below.
   Architecture. A modular pipeline consisting of five stages: item loading, generation, rollout archival, frame selection and extraction, and scoring. Generation is separated from scoring by the archive, so a metric change never requires regeneration, and per-task logic is confined to individual modules for each task.
   Frame selection. Tasks place their answer in different parts of the rollout, so the rule is set per task. Every task is additionally re-scored under oracle frame choice, separating readout failure from frame-selection failure and offering a best-case scenario metric result.
   Video fidelity. Three gates screen each rollout: scene preservation outside the region the model was asked to mark, instruction compliance, and temporal stability across selected frames. A threshold will be determined per gate, and rollouts failing these thresholds are recorded as invalid, and the invalid rate is itself reported.

Scoring and baselines. Metrics match each output convention, normalised to $[0, 1]$ and reported per task and per level, never as a single aggregate.

5. **Model evaluation.**
   The panel is drawn to overlap with the RoboTrustBench evaluation set: open-weight general-purpose generators (Wan2.2, HunyuanVideo-1.5), robot-pretrained world models (Cosmos-2B/14B, LingBot-World), and proprietary frontier models (Veo-3.1-Fast, Kling-v2.6). Runs are ordered by level of understanding, Scene first and Robot State last. Every rollout is archived with model version, sampler configuration, prompt hash, and seed.
6. **Results and analysis.**
   The archived rollouts are scored and written up, and the failure taxonomy is built.
   Reporting. Scores are reported per task and per level. Every score is accompanied by the invalid rate under the fidelity gates. Results are broken down three ways: by model family, by scenario type, and by rule-based against oracle frame selection, the last of which separates what the model rendered from what our frame selection rule recovered.
   Failure characterisation. A sample of rollouts is inspected and sorted into failure categories, with the category set developed on the pilot items and frozen before it is applied to the rest. The main distinction is between failures of understanding, where the model renders a coherent but wrong answer, and failures of expression, where the answer cannot be read at all: common errors would likely be ignored output conventions, altered or hallucinated scene geometry, or temporal instability. Category distributions are reported per model, task, and level.
7. **Writing and release.**
   The codebase is consolidated along the five pipeline stages defined in Phase 4 — item loading, generation, archival, frame selection and extraction, and scoring — with versions pinned, determinism verified end to end, and a single entrypoint reproducing any table from the manifest and archive. Documentation covers a quickstart, the per-task protocol, and an index mapping every reported number to its rollout ids. The release is the item set, prompt templates, extraction and scoring code, and the raw rollouts. The protocol is only usable if reproducible, and since generation is stochastic and closed models are versioned and mutable, the rollouts are the only durable record of what was measured.

4.2 Timeline

Figure 1 shows the schedule. The allocation is driven by three constraints. The first four phases each overlap their successor by a week, since data curation can begin while scoping concludes and the scoring protocol takes shape against evidence from the pilot; sequencing them strictly would waste time that the dependencies do not require.

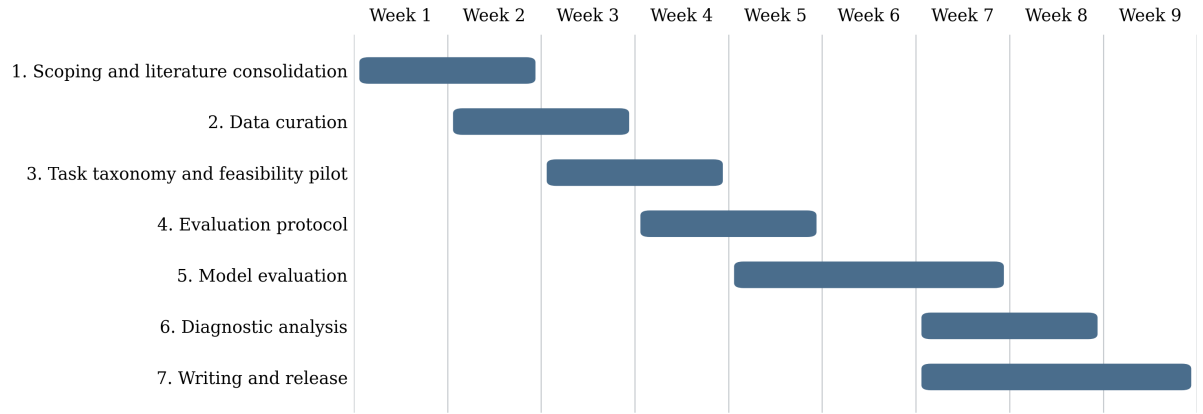


Figure 1: Tentative project schedule.

A Tables

Scenario	Main set	Share	Pilot
Normal	533	44.2%	9
Constraint-Sensitive	345	28.6%	6
Counterfactual	228	18.9%	3
Adversarial	101	8.4%	2
Total	1,207	100%	20

Table 1: Pilot subset, stratified by scenario type.

Level	**Task**	**Readout convention**	**Anchor**
1. Scene	T1 Semantic segmentation	Objects and robot embodiment each overlaid in a flat distinct colour, geometry unmodified	SAM 3
	T2 Depth perception	Scene re-rendered as a depth map under a stated colour convention	Depth Anything 3
2. Interaction	T3 Grasp region segmentation	Graspable region of the instruction target highlighted	Grasp specialist
	T4 Contact detection	Robot–object contact regions marked, other geometry unchanged	Gripper state, labels
	T5 Active object detection	Active object in one colour, unrelated objects in grey	Instruction, labels
3. Robot State	T6 Robot keypoint detection	Joint and end-effector keypoints marked on the arm	RoboPEPP
	T7 Robot pose estimation	Arm configuration rendered under a fixed pose convention	RoboPEPP

Table 2: Task taxonomy: the seven shortlisted readout tasks over three levels of understanding. Tiers are as defined in Phase 3.

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